

Cloud Warehouse Platform | Continue Claude Chat | WhatsApp | SIMATS Online Lab | Download file | iLovePDF | Ask Gemini | Relaunch to update | All Bookmarks

Not secure | simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=11

SIMATS
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

H Harshavardhan
1925117718D

QUESTIONS 4 / 5 completed

Heap Sort

HEAP SORT

Write a program that implements the Heap sort for the following list of elements:
23,56,12,34,78 and 29

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 void heapify(int arr[], int n, int i) {
4     int largest = i;
5     int left = 2 * i + 1;
6     int right = 2 * i + 2;
7
8     if (left < n && arr[left] > arr[largest])
9         largest = left;
10
11    if (right < n && arr[right] > arr[largest])
12        largest = right;
13
14    if (largest != i) {
15        int temp = arr[i];
16        arr[i] = arr[largest];
17        arr[largest] = temp;
18        heapify(arr, n, largest);
19    }
20 }
21 }
```

OUTPUT

Sorted array: 12 23 29 34 56 78

INPUT

23 56 12 34 78 29

Cloud Warehouse Platform | Continue Claude Chat | WhatsApp | SIMATS Online Lab | Download file | iLovePDF | Ask Gemini

Not secure | simatslab.saveetha.com/practice_app/onlineide/onlineide?topic=11 | Relaunch to update | All Bookmarks



SIMATS Online Programming Lab
DS PROGRAMMING

H Marshavardhan
19251177100

QUESTIONS - 4 / 5 completed

Heap Sort

HEAP SORT

Write a program that implements the Heap sort for the following list of elements :
23,56,12,34,78 and 29

PROGRAM EDITOR

C Run Save

```
20 }
21 }
22 void heapSort(int arr[], int n) {
23     for (int i = n / 2 - 1; i >= 0; i--)
24         heapify(arr, n, i);
25     for (int i = n - 1; i > 0; i--) {
26         int temp = arr[0];
27         arr[0] = arr[i];
28         arr[i] = temp;
29         heapify(arr, i, 0);
30     }
31 }
32 int main() {
33     int arr[] = {23, 56, 12, 34, 78, 29};
34     int n = 6;
35     heapSort(arr, n);
36     printf("Sorted array: ");
37     for (int i = 0; i < n; i++)
38         printf("%d ", arr[i]);
39     return 0;
40 }
```

OUTPUT

Sorted array: 12 23 29 34 56 78

INPUT

23 56 12 34 78 29